

Replicating Table 1, Panel B from Chernozhukov et al. (2017)

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May 27, 2026

1 The Original Paper

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, and Newey (2017), “Double/Debiased/Neyman Machine Learning of Treatment Effects,” *American Economic Review: Papers & Proceedings*, 107(5): 261–265, introduces the Double/Debiased Machine Learning (Double ML) estimator for causal inference in settings where high-dimensional nuisance parameters must be estimated. The central research question is how to obtain valid, root- N consistent estimates of a treatment effect parameter θ_0 when the confounding functions are estimated using flexible machine learning methods that introduce regularization bias.

The empirical strategy rests on two ingredients. First, the estimating equation is made Neyman-orthogonal, meaning it is locally insensitive to small errors in the nuisance parameter estimates. Second, cross-fitting is used to split the sample into folds, training the nuisance functions on held-out data to avoid overfitting bias. Together these two steps remove the regularization bias that would otherwise contaminate a naive plug-in estimator.

2 The Result of Interest

The specific result I replicate is the Mean Average Treatment Effect (ATE) of 401(k) eligibility on net total financial assets reported in Table 1, Panel B (Partially Linear Model, 2-fold cross-fitting), Ensemble column of the online appendix. The paper reports a Mean ATE of \$9,043 with a standard error of 1,432.

This result is the headline finding of the empirical illustration: after flexibly controlling for income and other confounders using an ensemble of machine learning methods, 401(k) eligibility raises net financial assets by roughly \$9,000. It is the cleanest single number to target because it corresponds to the default specification in the replication code (`Master.R`), uses the partially linear model which is the paper’s primary specification, and the Ensemble method which combines all ML approaches optimally.

3 Data and Methods

The data come from the 1991 Survey of Income and Program Participation (SIPP), the same dataset used in Chernozhukov and Hansen (2004). The sample contains 9,915 observations. The outcome variable `net_tfa` is net total financial assets (in USD), defined as the sum of IRA balances, 401(k) balances, checking accounts, saving bonds, and other interest-earning assets less non-mortgage debt. The treatment variable `e401` is a binary indicator for eligibility to enroll in a 401(k) plan. The nine covariates used in the model are age, income, years of education, family size, and binary indicators for married status, two-earner household, defined benefit pension, IRA participation, and home ownership.

`preprocess.R` loads the raw Stata file `sipp1991.dta` from `input/`, selects the eleven relevant variables, and writes a clean CSV to `temp/clean_data.csv`. `analysis.R` loads that file, sources the original helper functions `ML_Functions.R` and `Moment_Functions.R` from `input/`, and calls `DoubleML()` with the partially linear model specification and 2-fold cross-fitting. The estimation is repeated over 10 random sample splits (the paper uses 100) using parallel computation across 4 cores. The Mean ATE, Median ATE, and their standard errors are computed following the paper’s formulas exactly. Results are written to `output/tables/main_result.tex`, which is `\input{}`ed directly into this document.

4 Replication Results

Table 1: Replication of Table 1, Panel B (Partially Linear Model, 2-fold) from Chernozhukov et al. (2017)

	This Replication	Paper (Ensemble)
Mean ATE (\$)	9184.13	9,043
Std. Error	1401.53	1,432
95% CI	[6437.13, 11931.12]	—
Median ATE (\$)	9102.80	9,061
Median SE	1401.53	1,343

Note: Based on 10 sample splits, 2-fold cross-fitting.

Paper values from Table 1, Panel B, Ensemble column.

Our replication obtains a Mean ATE of \$9,184 with a standard error of 1,402, compared to the paper’s reported value of \$9,043 (s.e. 1,432). The difference is \$141, or approximately 1.6% of the paper’s estimate. This close match is reassuring and consistent with the theory: the Double ML estimator is designed to be robust to the choice of ML method, and indeed our Ensemble estimate lands very close to the paper’s despite using only 10 sample splits rather than 100. The slight discrepancy is expected — with fewer splits the mean estimate has higher sampling variance across replications. The standard errors are also very similar (\$1,402 vs \$1,432), suggesting the variance estimation procedure is working correctly.

5 What I Learned

The easiest part of the replication was the estimation itself — once the helper functions were correctly sourced and the `DoubleML()` call matched `Master.R` exactly, the code ran without modification. The hardest part was the path management: the original code assumed all files lived in the same working directory, while the Gentzkow-Shapiro structure separates `input/`, `code/`, and `temp/`. In particular, `Moment_Functions.R` internally sources `ML_Functions.R` with a bare filename, which required patching to use an absolute path. This is a common and underappreciated reproducibility failure — code that works perfectly in the author’s environment silently breaks in any other directory structure.

More broadly, this replication taught me that Double ML is remarkably stable: six very different machine learning methods (RLasso, Trees, Random Forest, Boosting, Neural Net, Ensemble) all produce estimates in the \$7,900–\$9,200 range, none statistically distinguishable from one another. This stability is exactly what the Neyman orthogonality theory predicts, and seeing it empirically is more convincing than any theorem. If I were the original authors, the one thing that would have made replication easier is absolute rather than relative paths in the helper files, and a brief note in the README specifying that all scripts must be run from the same directory as the data.

References

- [1] Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., and Newey, W. (2017). “Double/Debiased/Neyman Machine Learning of Treatment Effects.” *American Economic Review: Papers & Proceedings*, 107(5): 261–265.